

MYPADICARE



INTELLIGENT PADDY DISEASE DETECTION WITH LLM ENHANCED TREATMENT RECOMMENDATION



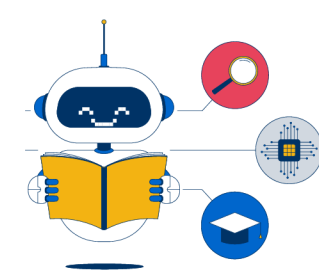
Introduction

- More than **50%** of the world feeds on paddy, but diseases decreases yield crop.
- Able to give **real-time disease detection and instant treatment guidance**, helping farmers protect crops efficiently and sustainably.
- Scalable for Malaysia and ASEAN, supporting the **National AI Strategy and SIRIM's mission** in agricultural innovation.



Objectives

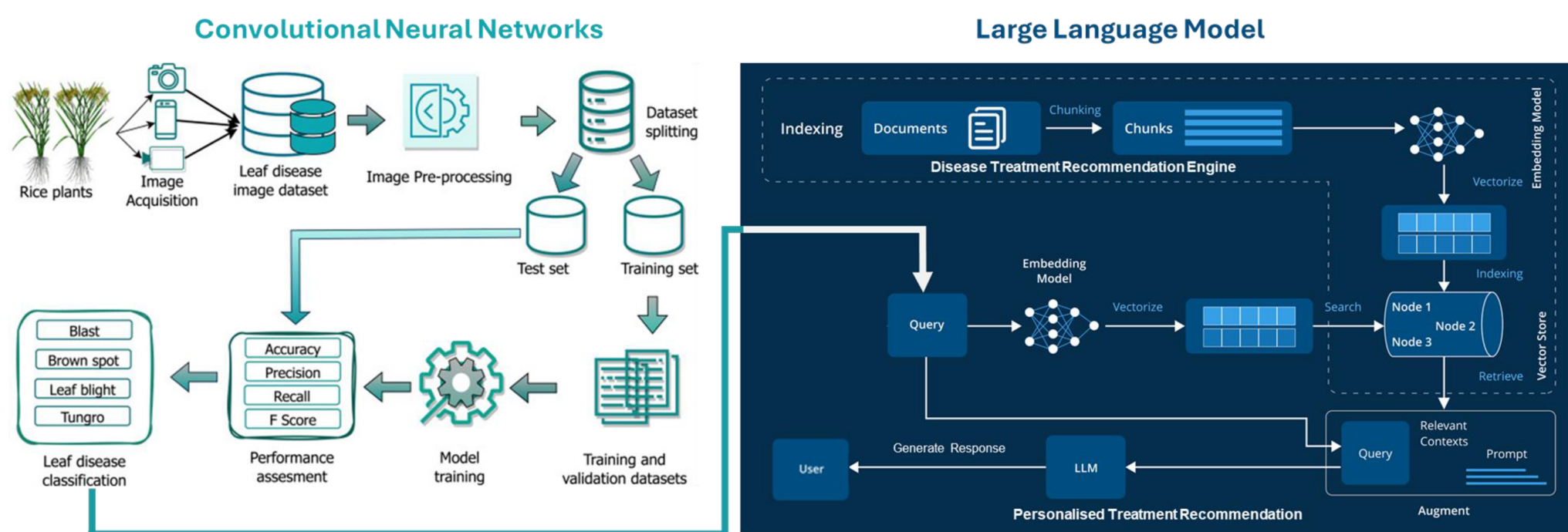
- Develop a **CNN-based model** for automated disease detection.
- Implement an **LLM-driven** system for personalized treatment recommendations.
- Deploy an accessible **mobile application** for farmers.



Problem Statement

- Lack of scalable solutions for real-time disease detection and treatment guidance while traditional detections are unreliable, slow and unaccurate,

Solution Framework



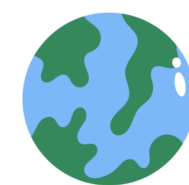
Value Propositions

- **High accuracy (90%)** using cross-validation and real-world testing for fast disease classification from uploaded images.
- Farmer-friendly explanations with multi-language support, powered by **Gemini API**.
- Mobile/Web app with a **simple upload feature, real-time results, and step-by-step treatment guides**.
- **Flexible business model:** pay-per-use, subscription, and supply-chain integration.



Impact & Benefits

- Early intervention **reduces crop losses** and optimizes input use.
- Automated recommendations **minimize pesticide misuse and environmental impact**.
- Platform fosters **data-driven, self-reliant** farming communities.
- Can reduce yield losses by up to **70%**



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